What if we had a Chinese-only LLM?

I used to assume that talking to an LLM in Chinese would take fewer tokens than talking to it in English. My reasoning was simple: Chinese editions of books run about 30% thinner than their English counterparts, largely because the language packs more information into each character — on average, a single Chinese character carries about as much meaning as a five-letter English word, and Chinese doesn't use spaces between words.

But "information density" isn't the same thing as "token efficiency" in an LLM. To put my intuition to the test, I picked six models — four developed by Chinese companies and two GPT models — and compared their vocabularies:

Model	Total weight parameters	Vocabulary size	Chinese entries	% (Chinese)	English entries	% (English)
Kimi K3.1 Pro	2.8T	163,840	69,449	42.45%	79,245	48.44%
Hy3	295B	120,832	45,441	37.87%	62,039	51.70%
DeepSeek V4.1	552B	129,280	35,335	27.61%	68,695	53.67%
GLM-5.3	744B	154,880	28,603	18.48%	95,412	61.63%
GPT-5 o200k	Not public	200,019	7,532	3.77%	122,491	61.25%
GPT-4 cl100k	Not public	100,277	840	0.84%	88,259	88.03%

Model size turns out not to matter much here. These models range from 295 billion to 2.8 trillion parameters — a tenfold spread — yet every vocabulary sits at roughly 150K tokens. It also makes sense that the Chinese models reserve a much larger share of their vocabularies for Chinese than GPT does, since they're trained on far more Chinese text.

I asked my agent to put together 30 short sentences and 70 long ones, all built from high-frequency expressions in everyday situations, while deliberately leaving out scientific and technical jargon. Each sentence came in a Chinese–English pair. The short ones look like this:

(1) The forecast says it will be sunny tomorrow.

(2) Is there any milk left in the fridge?

The long ones look like this:

(1) The kettle I bought last week makes a noise when it boils; customer support says I can return it within seven days, so I'll send it back tomorrow.

(2) My kid keeps losing points on word problems this term; when I went over the mistakes, I realized the real problem was reading the question.

(3) My fixed deposit matures next Wednesday; the teller asked whether I wanted to roll it over, but I'd rather withdraw part of it as an emergency fund.

Then I counted the tokens each model used on those sentences. The table below is based on the original wording of the examples:

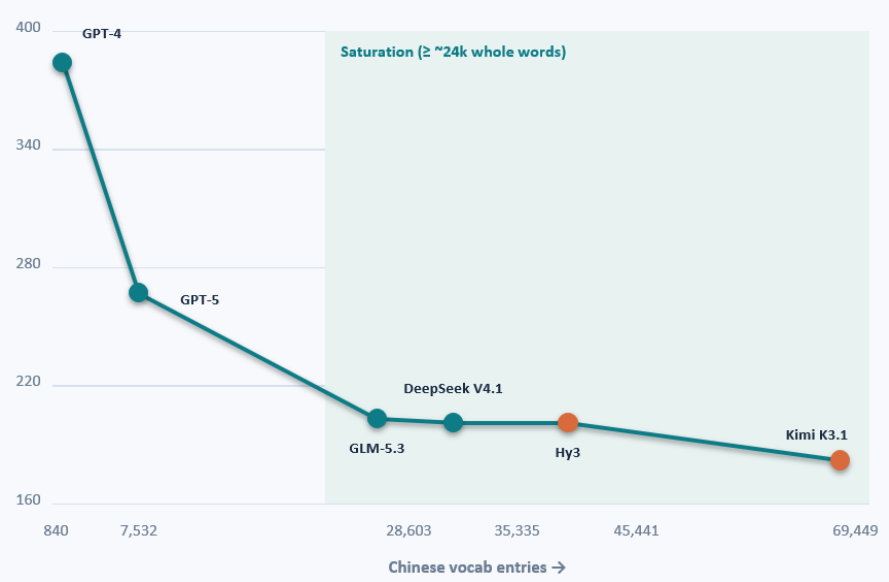
Model	Chinese %	Chinese tokens	English tokens	English / Chinese	Chinese saves
Kimi K3.1 Pro	42.5%	1,512	2,118	1.40	+28.6%
Hy3	37.9%	1,674	2,175	1.30	+23.0%
DeepSeek V4.1	27.6%	1,655	2,163	1.31	+23.5%
GLM-5.3	18.5%	1,687	2,160	1.28	+21.9%
GPT-5 o200k	3.8%	2,134	2,113	0.99	-1.0%
GPT-4 cl100k	0.8%	3,141	2,160	0.69	-45.4%

Three things stood out to me:

(1) For the same meaning, the Chinese sentences take fewer tokens — about 1,600 for all 100 sentences (30 short + 70 long) — compared with roughly 2,100 for their English counterparts.

(2) GPT-4 has only 840 Chinese entries, nowhere near enough to cover everyday expressions, which is why it comes out worst on Chinese token efficiency. GPT-5 has more than 7,500 Chinese entries and does much better.

(3) As long as a model has enough Chinese entries in its vocabulary, I'd expect Chinese to stay about 20% more token-efficient — at least outside scientific and technical writing. But how many entries is enough? Using the token counts from the 30 short sentences as a sample, I worked out the figure below: a vocabulary with around 25K Chinese entries is already close to saturation.



When I looked into the literature afterward, I found that 32K Chinese entries cover about 99% of everyday expressions.

That got me thinking: what if we trained a Chinese-only LLM with a 32K vocabulary? It would have three advantages:

（1） Fewer parameters: Qwen3.8-27B has 27 billion parameters and a 248K vocabulary. Shrink the vocabulary to 32K and the model drops to 24B parameters. I'll skip the detailed calculation here.

（2） Better token efficiency: as the table above shows, Chinese comes out about 20–30% ahead, which means fewer tokens for the same text.

（3） Faster decoding: a smaller vocabulary and fewer parameters mean the model can generate tokens faster.

Maybe it's time we actually built a Chinese-only LLM.